

VSEPR THEORY

JEE Advanced Preparation - Chemistry

Valence Shell Electron Pair Repulsion Theory

Topic Overview

VSEPR (Valence Shell Electron Pair Repulsion) Theory predicts molecular shapes based on the repulsion between electron pairs in the valence shell of the central atom. It is a simple, powerful model widely used in JEE Advanced questions.

1. POSTULATES OF VSEPR THEORY

- **Postulate 1:** Electron pairs in the valence shell of the central atom **repel each other** and arrange themselves to be as far apart as possible to minimise repulsion.
- **Postulate 2:** Both **bond pairs (bp)** and **lone pairs (lp)** occupy space and repel each other.
- **Postulate 3:** The **repulsion order** is: $\text{lp-lp} > \text{lp-bp} > \text{bp-bp}$ (lone pair–lone pair repulsion is maximum).
- **Postulate 4:** Lone pairs occupy more space than bond pairs because they are attracted by only one nucleus.
- **Postulate 5:** Multiple bonds (double or triple) are treated as a **single electron domain** for the purpose of geometry prediction.
- **Postulate 6:** The actual **molecular geometry** (shape) describes only the positions of atoms, not lone pairs.
- **Postulate 7:** Repulsion between electron pairs determines both bond angles and shape.

2. AX_nE_m NOTATION

Notation: AX_nE_m

- A = central atom
- X = bonded atoms (bond pairs), subscript n = number of bond pairs
- E = lone pairs on central atom, subscript m = number of lone pairs
- Total electron domains = $n + m$

Example: NH₃ is AX₃E₁ (3 bond pairs + 1 lone pair = 4 total domains → tetrahedral electron geometry, trigonal pyramidal molecular shape)

3. REPULSION ORDER AND LONE PAIR EFFECTS

The strength of repulsion follows: **lp-lp** > **lp-bp** > **bp-bp**. Lone pairs are "fatter" — they spread over more space and push bond pairs closer together, reducing bond angles.

Why NH₃ has a bond angle of 107° (less than 109.5°):

- NH₃ has 3 bond pairs + 1 lone pair (AX₃E₁)
- Lone pair repels the 3 bond pairs more strongly than bond pairs repel each other
- This compresses the H-N-H angles from 109.5° to 107°

Why H₂O has a bond angle of 104.5° (less than NH₃):

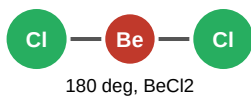
- H₂O has 2 bond pairs + 2 lone pairs (AX₂E₂)
- Two lone pairs cause even stronger compression of H-O-H angles to 104.5°
- Bond angle order: CH₄ (109.5°) > NH₃ (107°) > H₂O (104.5°)

4. VSEPR GEOMETRY TABLE

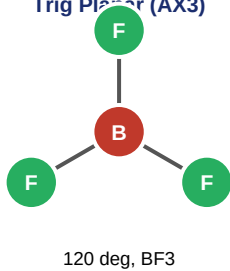
Formula	Domains	Shape	Bond Angle	Example
AX ₂	2	Linear	180°	BeCl ₂ , CO ₂
AX ₂ E	3	Bent (V-shaped)	<120°	SO ₂ , SnCl ₂
AX ₃	3	Trigonal Planar	120°	BF ₃ , AlCl ₃
AX ₃ E	4	Trigonal Pyramidal	107°	NH ₃ , PCl ₃
AX ₂ E ₂	4	Bent (V-shaped)	104.5°	H ₂ O, OF ₂
AX ₄	4	Tetrahedral	109.5°	CH ₄ , CCl ₄
AX ₄ E	5	See-saw	<90°/ <120°	SF ₄
AX ₃ E ₂	5	T-shaped	<90°	ClF ₃ , BrF ₃
AX ₂ E ₃	5	Linear	180°	XeF ₂ , I ₃ -
AX ₅	5	Trig Bipyramidal	90°/120°	PCl ₅
AX ₅ E	6	Square Pyramidal	<90°	IF ₅ , BrF ₅
AX ₄ E ₂	6	Square Planar	90°	XeF ₄
AX ₆	6	Octahedral	90°	SF ₆ , PCl ₆ -

5. MOLECULAR GEOMETRY DIAGRAMS

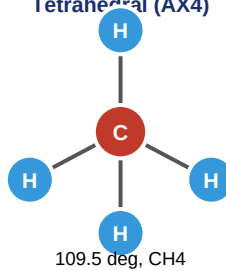
Linear (AX2)



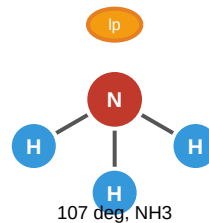
Trig Planar (AX3)



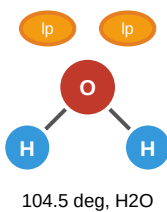
Tetrahedral (AX4)



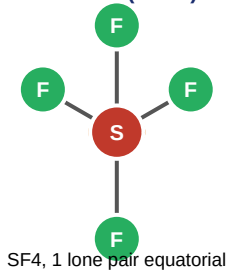
Trig Pyramidal (AX3E)



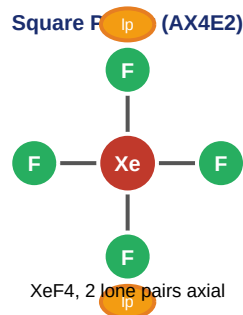
Bent (AX2E2)



See-saw (AX4E)



Square Planar (AX4E2)



6. WORKED EXAMPLES

Example 1 — Shape of SO_3^{2-} (sulfite ion)

Question: Predict the shape and bond angle of SO_3^{2-} .

Solution:

- Central atom: S; Valence electrons of S = 6; Charge = -2
- Total valence electrons: $\text{S}(6) + 3 \times \text{O}(6) + 2(\text{charge}) = 26$
- Lewis structure: S has 3 S-O bonds + 1 lone pair on S
- Electron domains on S: 3 bond pairs + 1 lone pair = 4 total $\rightarrow \text{AX}_3\text{E}_1$
- Electron geometry: Tetrahedral
- Molecular shape: Trigonal Pyramidal
- Bond angle: slightly less than 109.5° (due to lone pair) $\approx 106^\circ$
- Answer: SO_3^{2-} is AX_3E_1 , Trigonal Pyramidal, $\sim 106^\circ$

Example 2 — Shape of XeOF₄

Question: Predict the shape of XeOF₄.

Solution:

- Central atom: Xe; Valence electrons = 8
- Bonds: 4 F atoms (4 bond pairs) + 1 O atom (1 double bond = 1 domain) = 5 bond domains
- Lone pairs on Xe: $(8 - 5 \times 2)/2$... actually: 8 valence e⁻ on Xe, 5 bonds use 5, lone pairs = $(8-5)=$ need counting
- Better: total electron domains = 5 bonds + 1 lone pair = 6 → AX₅E₁ → octahedral electron geometry
- Molecular shape: Square Pyramidal (1 lone pair occupies one octahedral position)
- Bond angles: F-Xe-F $\approx 90^\circ$; Xe=O bond slightly shorter
- Answer: XeOF₄ is AX₅E₁, Square Pyramidal

Example 3 — Shape of PCl₄⁺ (phosphonium-like cation)

Question: Predict the shape of PCl₄⁺.

Solution:

- Central atom: P; Valence electrons = 5; Charge = +1, so remove 1 electron → 4 electrons
- Bonds: 4 Cl atoms → 4 bond pairs
- Total electron domains on P: $(5-1)/2$... use H formula: $H = (5+4-1)/2 = 4$
- Actually: PCl₄⁺ has 4 bond pairs + 0 lone pairs on P (positive charge removes lone pair)
- AX₄ → Tetrahedral
- Bond angle: 109.5°
- Answer: PCl₄⁺ is AX₄, Tetrahedral, 109.5°

Example 4 — Shape of ICl_4^- (tetrachloroiodate)

Question: Predict the shape of ICl_4^- .

Solution:

- Central atom: I; Valence electrons = 7; Charge = -1 $\rightarrow$ total = 8 electrons
- Bonds: 4 Cl atoms $\rightarrow$ 4 bond pairs
- Using hybridisation: $H = (7+4+1)/2 = 6 \rightarrow \text{sp}^3\text{d}^2$
- Total domains: 4 bond pairs + 2 lone pairs = 6 $\rightarrow \text{AX}_4\text{E}_2$
- Electron geometry: Octahedral
- 2 lone pairs occupy trans (opposite/axial) positions to minimise lp-lp repulsion
- Molecular shape: Square Planar
- Bond angle: 90°
- Answer: ICl_4^- is AX_4E_2 , Square Planar, 90°

Example 5 — Why is bond angle in ClF_3 less than 90° ?

Question: ClF_3 has T-shaped geometry. Explain why its F-Cl-F angles are less than 90° .

Solution:

- ClF_3 : Cl has 7 valence electrons; 3 F atoms $\rightarrow \text{AX}_3\text{E}_2$
- 5 total electron domains (sp^3d hybridisation)
- T-shaped geometry: 3 bond pairs + 2 lone pairs in equatorial positions
- The 2 lone pairs occupy equatorial positions (less repulsion than if in axial)
- These lone pairs repel the 2 axial F atoms, compressing the F(axial)-Cl-F(equatorial) angles
- Expected angle 90° , but lp-bp repulsion reduces it to $\sim 87.5^\circ$
- Answer: lp-bp repulsion from 2 equatorial lone pairs compresses bond angles below 90°

7. KEY FACTS — QUICK REVISION

KEY FACTS / QUICK REVISION — VSEPR THEORY

- Repulsion order: $\text{lp-lp} > \text{lp-bp} > \text{bp-bp}$
- Lone pairs occupy more space $\rightarrow$ compress bond angles
- CH_4 (109.5°) $>$ NH_3 (107°) $>$ H_2O (104.5°) — each lone pair reduces angle
- Multiple bonds treated as single electron domain in VSEPR
- In trigonal bipyramidal, lone pairs prefer EQUATORIAL positions
- In octahedral, lone pairs prefer TRANS (axial) positions
- T-shaped = AX_3E_2 (sp^3d); Square Planar = AX_4E_2 (sp^3d^2)
- XeF_2 linear (3 lp equatorial); XeF_4 square planar (2 lp axial)
- AX_2E (bent $\text{SO}_2 \sim 119^\circ$) vs AX_2E_2 (bent $\text{H}_2\text{O} 104.5^\circ$)
- PCl_4^+ is tetrahedral; ICl_4^- is square planar

8. PRACTICE QUESTIONS — JEE ADVANCED LEVEL

1. The VSEPR notation for XeF_4 is:

- (A) AX_4E
- (B) AX_4E_2
- (C) AX_4E_3
- (D) AX_6

2. Which molecule has the smallest bond angle?

- (A) NH_3
- (B) H_2O
- (C) CH_4
- (D) BF_3

3. In which species do lone pairs occupy equatorial positions?

- (A) XeF_4
- (B) SF_4
- (C) ICl_4^-
- (D) PCl_5

4. The shape of SO_2 is best described as:

- (A) Linear
- (B) Trigonal Planar
- (C) Bent (V-shaped)
- (D) Tetrahedral

5. ClF_3 has T-shaped geometry. The VSEPR notation is:

- (A) AX_3E
- (B) AX_3E_2
- (C) AX_3E_3
- (D) AX_4E

6. Which of the following pairs of molecules have the same shape?

- (A) NH_3 and BF_3
- (B) H_2O and CO_2
- (C) XeF_2 and CO_2
- (D) SF_4 and CH_4

7. The bond angle in water is less than in ammonia because:

- (A) O-H bond is longer
- (B) H_2O has more lone pairs
- (C) O is less electronegative
- (D) H_2O is linear

8. IF_5 has which molecular geometry?

- (A) Trigonal bipyramidal
- (B) Square pyramidal
- (C) Octahedral
- (D) Pentagonal bipyramidal

9. Which molecule has a zero net dipole moment due to its geometry?

(A) NH_3

(B) H_2O

(C) BF_3

(D) SO_2

10. The correct order of bond angles is:

(A) $\text{H}_2\text{O} < \text{NH}_3 < \text{CH}_4$

(B) $\text{NH}_3 < \text{H}_2\text{O} < \text{CH}_4$

(C) $\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$

(D) $\text{H}_2\text{O} < \text{CH}_4 < \text{NH}_3$

9. ANSWER KEY WITH EXPLANATIONS

1. Answer: B — AX₄E₂

XeF₄: Xe has 8 valence e⁻, 4 F atoms → 4 bond pairs; remaining 4 electrons = 2 lone pairs. AX₄E₂ → 6 total domains → sp³d² → Square Planar.

2. Answer: B — H₂O

H₂O has 2 lone pairs (AX₂E₂), giving maximum lone pair compression → 104.5°. NH₃ has 107°; CH₄ has 109.5°; BF₃ has 120°.

3. Answer: B — SF₄

SF₄ is AX₄E in sp³d (trigonal bipyramidal). The single lone pair occupies an equatorial position. XeF₄ and ICl₄⁻ have lone pairs in axial positions.

4. Answer: C — Bent

SO₂: S has 3 electron domains (AX₂E₁) → trigonal planar electron geometry → bent molecular geometry ≈ 119°.

5. Answer: B — AX₃E₂

ClF₃: 5 electron domains (3 bp + 2 lp) → AX₃E₂ → T-shaped.

6. Answer: C — XeF₂ and CO₂

Both are linear: XeF₂ (AX₂E₃, 3 lone pairs equatorial, 2 axial bonds) and CO₂ (AX₂, no lone pairs). Both molecules have linear geometry.

7. Answer: B

H₂O has 2 lone pairs vs NH₃'s 1 lone pair. More lone pairs = more lp-bp repulsion = smaller bond angle.

8. Answer: B — Square Pyramidal

IF₅: I has 7 valence e⁻, 5 F bonds → AX₅E₁ → 6 electron domains → octahedral electron geometry → square pyramidal molecular shape.

9. Answer: C — BF₃

BF₃ is trigonal planar (AX₃, 120°). The 3 B-F bond dipoles are symmetrically arranged → net dipole = 0. NH₃, H₂O, and SO₂ all have net dipole moments.

10. Answer: A — H₂O < NH₃ < CH₄

Lone pairs compress bond angles: CH₄ (no lp) = 109.5°; NH₃ (1 lp) = 107°; H₂O (2 lp) = 104.5°. So order is H₂O < NH₃ < CH₄.